Course Code	Course Title	L	T	P	C
MACSE519	**Machine Learning and Applications**	**3**	**0**	**2**	**4**
Pre-requisite	Nil	**Syllabus Version**			
		1			

Course Objectives

1. To deliver insights into the foundational and mathematical principles of machine learning algorithms.
2. To impart skills to select and apply appropriate machine learning models to address real-world problems across diverse domains.
3. To explore emerging trends in machine learning and its advancements emphasizing societal impact and ethical considerations

Course Outcomes

1. Comprehend the fundamental properties of machine learning techniques.
2. Design and Develop supervised learning techniques for real-world data challenges.
3. Evaluate ensemble methods to improve model accuracy and robustness.
4. Develop an appropriate unsupervised learning model to manage unknown patterns.
5. Apply Reinforcement Learning concepts and algorithms for intelligent system design.

Module:1	**Introduction**	**6 hours**

Overview of Machine Learning (ML) Concepts - Applications - Types of ML algorithms - Hypothesis space and Inductive Bias - Data - Data handling - Model Selection and Evaluation: Train test split and cross-validation - Evaluation metrics

Module:2	**Supervised Learning**	**10 hours**

Linear Regression: Simple and Multiple Linear regression - Evaluating regression fit – Non-Linear Regression – Applications: Predicting Sales, Predicting price of Real Estate, Predicting Concrete Compression Strength - Multi-Class and Multi-Label classification -Decision tree learning: Decision tree representation - Decision tree algorithms- Hypothesis space search in decision tree learning - Inductive bias in decision tree learning - Instance based Learning: K nearest neighbor - Support Vector Machine (SVM) - Dual formulation - Maximum margin with noise - Nonlinear SVM and Kernel function.

Module:3	**Probabilistic and Ensemble Learning**	**9 hours**

Probability and Bayes Learning: Logistic Regression - Bayesian Learning, Naïve Bayes - Bayesian Belief Networks - Artificial Neural Networks (ANN): Biological motivation - ANN representation - Perceptron - Multilayer perceptron and the back propagation algorithm - Ensembles: Bias-Variance Tradeoff – Bagging - Random

Forest, Boosting – Adaboost - Applications: Predicting Fraud Insurance Claims, Email Spam filtering		
Module:4	**Unsupervised Learning**	**10 hours**
Clustering- Partitional Clustering: k-means - k-medoids - Hierarchical Clustering: Agglomerative - Divisive - Density based clustering: DBSCAN-Spectral clustering - Dimensionality Reduction: Principal Component Analysis - Linear Discriminant Analysis - Expectation Maximization - Mixture of Gaussians- Outliers- Anomaly Detection – Applications :Document Clustering - Customer Segmentation		
Module:5	**Advanced Machine Learning Concepts**	**8 hours**
Self-supervised Learning - Reinforcement Learning (RL) - RL Framework - Markov Decision Process - Policy - Planning algorithms - Q Learning algorithm - Online learning algorithms : Winnow algorithm - On-line to batch conversion - Overview of MLOps - Explainability in ML - Ethics in ML.		
Module:6	**Contemporary Issues**	**2 hours**
	Total Lecture Hours:	**45 hours**

Text Book(s)

Mehryar Mohri, Afshin Rostamizadeh, Ameet Talwalkar , "**Foundations of Machine Learning**", MIT Press, 2nd Edition, **2018**
Ethem Alpaydin, "**Introduction to Machine Learning**", MIT Press Prentice Hall of India, 4th Edition, **2020**

Reference Books

Tom Mitchell, "**Machine Learning**", McGraw Hill, 3rd Edition, **1997**
Shai Shalev-Shwartz and Shai Ben-David, "**Understanding Machine Learning: From Theory to Algorithm**", Cambridge University Press, **2014**
Stephen Marsland , "**Machine Learning: An Algorithmic Perspective**", Chapman and Hall CRC press, 2nd Edition, **2015**
Danish Haroon, "**Python Machine Learning Case Studies: Five Case Studies for the Data Scientist**", Apress, **2017**

Indicative Experiments

1. To effectively **predict alumni donation likelihood**, the university employs a comprehensive preprocessing strategy on its dataset. Numerical features such as annual income (in lakhs) and previous donation amounts (in rupees) are scaled to ensure all features contribute equally to the model. Categorical variables including employment sector, degree program, and marital status are encoded into numerical representations suitable for machine learning algorithms. Furthermore, the geographical location of alumni is transformed, potentially into binary features indicating specific regions or through one-hot encoding to capture more granular location information. These	2 hours

meticulous preprocessing steps are crucial for preparing the data for robust model training and ultimately enhancing the accuracy of predicting which alumni are most likely to donate.	
2. Use **linear regression** to predict a student's final academic performance based on features such as attendance percentage, internal assessment scores, assignment submissions, project grades, previous semester GPA, participation in extracurricular activities, study hours per week, socio-economic background, number of courses taken in the semester, involvement in academic clubs, time spent on campus, previous academic performance, and faculty feedback ratings.	2 hours
3. You have Computational Fluid Dynamics (CFD) simulation results for turbine blades under varying loading and flow conditions. Each simulation provides features like maximum principal stress, Von Mises stress, maximum temperature, displacement magnitude, surface pressure, fatigue life estimate, Reynolds number, and wall shear stress. Using this data, **formulate a decision tree approach** to classify whether a blade is likely to "Fail" or remain "Safe."	2 hours
4. Use the **KNN classifier** to predict the biochemical composition of a sample into enzyme concentration levels (e.g., Low, Medium, High) based on features such as pH level, enzyme activity, substrate concentration, temperature, protein concentration, absorbance at specific wavelengths, reaction time, catalyst concentration, substrate type, ion concentration, viscosity of the solution, and absorbance at 280 nm	2 hours
5. Apply **Support Vector Machines (SVM)** to predict whether a university student will stay enrolled or drop out based on features such as high school GPA, entrance exam score, extracurricular activities, letter of recommendation score, work experience, number of club activities, financial aid status, number of failed courses, mental health status, attendance percentage, family support, and work-study participation.	2 hours
6. **Implement logistic regression** to predict whether a hotel guest will cancel their booking based on features such as lead time, previous cancellations, deposit type (No Deposit, Non-Refundable, Refundable), customer type (Transient, Group, Contract, Transient-Party), number of special requests, average daily rate (ADR), reserved room type, total nights stayed, booking changes, market segment (Direct, Corporate, Online Travel Agent), assigned room type, repeated guest status, and distribution channel.	2 hours
7.	2 hours

A student at a multimedia school has created a web design project, which includes an animated logo, interactive media elements, and is designed using HTML/CSS. You need to predict whether this project will meet the design standards of a client and industry expectations based on the following features: **Features**: **Number of Pages Designed** (numeric): The number of pages in the website design. **Animation Complexity** (categorical): The complexity of the animation used (e.g., simple, moderate, complex). **Client Feedback Rating** (numeric): A rating (1-10 scale) provided by the client after previewing the project. **Interactive Features Implemented** (categorical): Whether the website includes interactive media elements like hover effects, videos, or dynamic content (yes/no). **HTML/CSS Coding Quality** (categorical): The quality of the HTML/CSS code written (e.g., high, medium, low). **Use of Multimedia Tools** (categorical): The tools used to create the multimedia content (e.g., Adobe After Effects, Blender, Final Cut Pro). **Project Deadline Met** (binary): Whether the project was completed on time (1 = yes, 0 = no). **Client Requirements Met** (binary): Whether the project met all client requirements (1 = yes, 0 = no). **Target Variable**: **Project Success** (binary): Whether the project is deemed successful (1 = success, 0 = failure) Using **Naive Bayes classification**, train a model to predict whether a multimedia project will be successful or unsuccessful based on the provided features. Evaluate the model's performance using accuracy, precision, recall, and F1-score	
8. Apply **K-Means clustering** to categorize cricket players based on their performance metrics using features such as batting average, strike rate, number of boundaries per match, total runs, bowling average, economy rate, number of wickets taken, dot ball percentage, fielding efficiency, and match impact score. The clustering can help group players into roles like power hitters, all-rounders, specialist bowlers, consistent batsmen, or fielding specialists.	2 hours
9.	2 hours

Apply DBSCAN clustering to categorize building construction projects based on features such as total construction cost, project duration, number of labor hours, material cost percentage, number of floors, safety incident count, construction method (traditional, modular, prefab), site complexity score, weather delays, and energy efficiency rating, to identify clusters of typical, high-risk, or high-efficiency construction projects.	
10. Use **Principal Component Analysis (PCA)** to analyze and reduce the dimensionality of participant and event data collected during a university techfest, based on features such as number of events participated, event types (coding, robotics, design, management), total scores earned, feedback ratings, number of workshops attended, team size, project submissions, number of prizes won, volunteer hours, and previous techfest experience.	2 hours
11. In the biochemical lab, researchers often run multi-step experiments like protein purification, DNA extraction, or enzymatic reactions. The experimental results depend heavily on parameters like temperature, pH, reaction time, reagent concentration and mixing speed Rather than manually tuning these parameters, you can use **Reinforcement Learning** to learn an optimal sequence of experimental settings that maximizes the experiment's success rate (e.g., purity of protein, yield of DNA, or enzymatic efficiency).	2 hours
12. Implement a **Random Forest classifier** to predict whether a patient is likely to have Tuberculosis (TB) based on clinical and diagnostic features. Use explainability techniques such as feature importance and individual prediction interpretation to ensure the model's decisions are understandable and medically relevant. **Features to include:** ● Chest X-ray abnormalities ● Duration of cough ● Weight loss ● Night sweats	2 hours

• Fever • Sputum test results • Previous TB history	
Total Laboratory Hours:	**30 hours**

Text Book(s)
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Mode of Evaluation : Continuous Assessment Test, Digital Assignment, Quiz, Final Assessment Test, Lab Continuous Assessment, Lab Final Assessment, Seminar

Recommended by Board of Studies :	23-05-2025
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